

Real-Time Oncology Review

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Purpose of the RTOR

The Oncology Center of Excellence Real-Time Oncology Review (RTOR) aims to provide a more efficient review process to ensure that safe and effective treatments are available to patients as early as possible, while improving review quality and engaging in early iterative communication with the applicant.

RTOR facilitates earlier submission of topline efficacy and safety results, prior to the submission of the complete application, to support an earlier start to the FDA's evaluation of the application.

Scope

Oncology drug applications, including New Drug Applications (NDAs) for new molecular entities (NMEs) submitted under section 505(b) of the Federal Food, Drug, and Cosmetic Act (FDCA), original Biologics License Applications (BLAs) submitted under section 351(a) of the Public Health Service (PHS) Act, and supplemental NDAs and BLAs, may be eligible for RTOR. Acceptance into the RTOR program does not guarantee or influence

approvability of the application, which is subject to the same statutory and regulatory requirements for approval as applications that are not included in RTOR. Participation by the applicant in this program is voluntary.

It is not the intent of this program to enable an early action prior to the specified user fee goal date. Although early approvals have occurred with RTOR applications, this may not be feasible for all applications due to specific issues that may be identified with the application or overall workload considerations.

Submissions to be considered for the RTOR program should meet the following criteria:

- Drugs likely to demonstrate substantial improvements over available therapy or meeting criteria for Expedited Programs.
- Straightforward study designs.
- Endpoints that can be easily interpreted (e.g., overall survival, response rates, etc.).

FDA will also consider whether adequate dose optimization has been performed to support the proposed dosage. Submissions with greater complexity will be considered for RTOR on a case by case basis. Additionally, FDA prefers the [Assessment Aid \(AAid\)](https://www.fda.gov/about-fda/oncology-center-excellence/assessment-aid) ([/about-fda/oncology-center-excellence/assessment-aid](https://www.fda.gov/about-fda/oncology-center-excellence/assessment-aid)) be used with RTOR.

Guidance

FDA issued a final guidance document, "[Real-Time Oncology Review \(RTOR\) Guidance for Industry](https://www.fda.gov/regulatory-information/search-fda-guidance-documents/real-time-oncology-review-rtor) ([/regulatory-information/search-fda-guidance-documents/real-time-oncology-review-rtor](https://www.fda.gov/regulatory-information/search-fda-guidance-documents/real-time-oncology-review-rtor))," in November 2023, to provide recommendations to applicants on the process for submission of selected new drug applications (NDAs) and biologics license applications (BLAs) with oncology indications for review under RTOR.

Standard Operating Procedures

Please note that these milestones are provided as a general guideline and may vary substantially across applications. The review team is expected to look at the pre-submission materials when they are submitted and to discuss their findings with the Cross-Discipline Team Leader (CDTL) and Division Director.

1. Week 0-3: At the time of topline results of a pivotal trial(s), if the eligibility criteria above is met, an applicant can apply for the RTOR program by contacting the appropriate application regulatory project manager (RPM) to request a copy of the RTOR Request table. Once the completed table has been submitted by the applicant, the clinical division director/deputy director, with input from the review team (including reviewers, team leaders, and management from all relevant review disciplines, and CDRH as applicable), will decide whether the application is eligible for the RTOR program. This decision will generally be made in approximately 20 business days of receipt of the completed RTOR Request Table and will be communicated to the applicant via email correspondence.
2. Week 3-6: Once an application is selected, a teleconference with the applicant may be scheduled, if necessary. The clinical division director/deputy director, the review team, and OCE staff may participate in this meeting. If the drug product is co-developed with a companion diagnostic, the diagnostic partner and CDRH should also be on the teleconference. If the application is not accepted into the RTOR program, the applicant should follow routine application submission procedures. Otherwise, FDA and the applicant will discuss the plan for RTOR and reach tentative agreement on proposed pre-submission timelines for the drug application. At this time, RTOR is only applicable to drug applications submitted to CDER and not applicable for the device applications submitted to CDRH.
3. Week 6-9: Under the RTOR program, the applicant would officially submit components of the marketing application as soon as they become available, bundled

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into a maximum of three pre-submissions and a final submission. An updated Reviewer’s Guide (refer to FDA’s technical specifications document: [eCTD Technical Conformance Guide \(/media/93818/download?attachment\)\)](#) which provides the eCTD location, submission date, and relevant hyperlinks to each pre-submission component, should be submitted with each pre-submission and the final submission of the complete application. When submitting complete ADaM datasets for key efficacy and safety tables/figures for the pivotal study, please refer to [OOD Safety standard data specifications \(/media/133252/download\)](#) for the requested format of safety datasets.

An example RTOR submission timeline can be found below.

Pre-Submission 1	Pre-Submission 2	Pre-Submission 3	Final Submission
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| <ul style="list-style-type: none"> ◦ User fee, if applicable ◦ Top line efficacy/safety tables/figures ◦ The protocol(s) and amendments (a list of major changes for each amendment), Statistical Analysis Plan(s) (SAP), and Data Monitoring Committee (DMC) charter(s) and DMC minutes ◦ All CMC information including list of all manufacturing, testing and critical intermediate facilities with addresses and FDA Establishment Identifier (FEI) numbers other than stability data for registration batches (if not available) for drug substance(s), drug product ◦ Case report forms (CRFs) ◦ Complete ADaM datasets for key efficacy and safety tables/figures for pivotal study(ies) (see OOD data specifications) | <ul style="list-style-type: none"> ◦ Key results, analysis, and datasets for other disciplines, if applicable ◦ Summary of data and rationales supporting dose and dosing regimen selection (including key population pharmacokinetics (PK), physiologically-based (PB)/PK and exposure-response reports, analyses programs, and datasets) ◦ Pediatric study plan (as required) ◦ Request for proprietary name review (as required) | <ul style="list-style-type: none"> ◦ Final study reports for all supportive studies, including pharmacology and toxicology studies ◦ Summary of clinical pharmacology studies and datasets supporting the conclusions | <ul style="list-style-type: none"> ◦ Completed AAid ◦ Final clinical study report(s) ◦ Complete (s)NDA/(s)BLA |
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(</media/133252/download?attachment>) for requested format of safety datasets)

- Statistical (e.g., SAS) programs
- Complete SDTM dataset package
- Proposed labeling

4. Week 10-16: Pre-submission meeting:

In addition to responding to the applicant's questions in the meeting package, FDA may share with the applicant preliminary key review questions or issues and critical analyses needed. If FDA requests additional analyses, the applicant may submit them before or at the time of submission of the marketing application. In some cases, if FDA agrees, the applicant may submit the requested additional analyses after the marketing application is submitted.

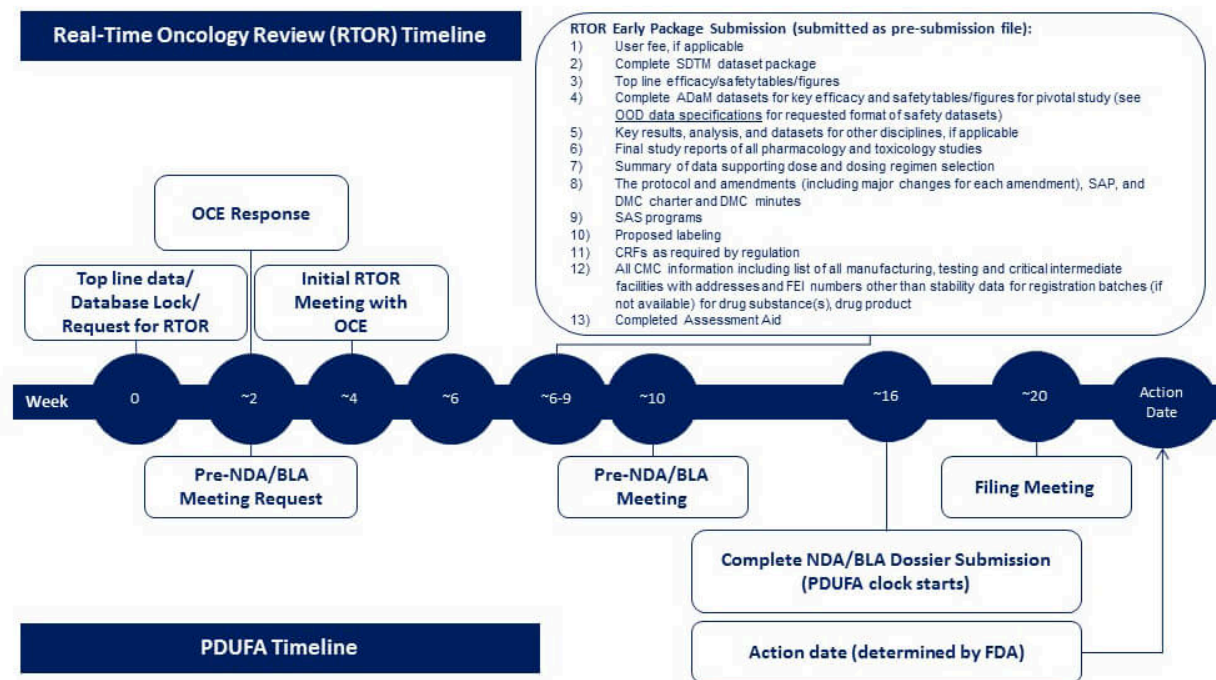
These discussions may be documented in the meeting minutes under the section, "Discussion of the Content of a Complete Application" for NDA NMEs or original 351(a) BLAs, under a new "Additional Items discussed" which can be added by the RPM for other applications or under specific questions as appropriate. For supplemental applications, RPMs may capture agreements discussed under "Additional Items Discussed" or under discussion of specific questions as appropriate in the official meeting minutes

5. Week 16-22: The applicant submits the complete marketing application. Once FDA receives the completed application, the review clock will start. The complete

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application will include any remaining components not previously submitted by the applicant.

Figure 1. General RTOR Timeline



* The diagram reflects a general timeline suggestion. Companies should discuss with FDA their submission proposal.

RTOR Frequently Asked Questions

1. What are the main differences between this and how the FDA currently treats NDAs and BLAs?

RTOR allows the FDA to conduct a preliminary review of the data from the pivotal trial(s) earlier in the review process before the applicant formally submits the complete application. First, the applicant will provide topline efficacy and safety data from the pivotal trial(s) along with justification for the FDA to determine whether RTOR would be appropriate for the application. If the FDA determines RTOR is an appropriate review plan and an agreed upon pre-submission timeline is reached with the review division, the applicant can start sending pre-submission data to the FDA 2-4 weeks after all patient data has been entered and locked by the applicant in their database and the applicant decides to request FDA approval. This pre-submission package should include key raw and derived (ADaM) datasets, including safety and efficacy tables and figures, the study protocol(s) and amendments, and a draft of the prescribing information. In addition, the applicant should also submit key results, analysis, and datasets for other disciplines (e.g., clinical pharmacology), if applicable. The FDA will start evaluating the pre-submitted data for sufficiency and integrity. This informed pre-analysis gives FDA reviewers an early opportunity to identify data quality and potential review issues and potentially provide early feedback to the applicant. RTOR is different from the mechanisms for rolling review in which, generally, complete modules (e.g., the complete clinical module) are submitted prior to a complete application submission

2. If not discussed at a pre-submission meeting, how and where should requests for participation in RTOR be submitted?

If interested in participating in this program, we advise the sponsor to contact the respective application RPM and request a copy of the RTOR Request table. The completed table, which includes information such as the justification for RTOR consideration and a proposed pre-submission timeline, should be submitted officially to the IND or NDA/BLA file.

3. How will we be notified of acceptance/rejection?

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For requests that are not discussed at a pre-submission meeting, a decision will generally be made in approximately 20 business days of receipt of the completed RTOR Request table. The FDA may schedule a teleconference between the sponsor and FDA review team to discuss the decision if necessary.

4. If accepted into the RTOR program, when are user fees due?

For supplemental applications, user fees are not currently being collected. For new molecular entities and original 351(a) biologics, PDUFA fees (when applicable) are due when the first component of the RTOR application is submitted (see [FDA User Fee Programs \(/industry/fda-user-fee-programs\)](/industry/fda-user-fee-programs)).

5. Does this impact PDUFA timelines?

PDUFA timelines will not be affected.

6. Does the division need to acknowledge the receipt of pre-submission RTOR submissions?

No. However, if the applicant requests a confirmation of receipt, then the application RPM can send a courtesy email confirming receipt.

7. What about companion diagnostic products?

Eligibility of applications with a companion diagnostic for RTOR will be determined on a case by case basis. At this time, RTOR is only applicable to oncology drug applications submitted to CDER and not applicable for device applications submitted to CDRH.

For Further Information

- [Assessment Aid \(/about-fda/oncology-center-excellence/assessment-aid\)](/about-fda/oncology-center-excellence/assessment-aid)

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- [Office of Oncologic Diseases \(OOD\) \(/about-fda/cder-offices-and-divisions/office-oncologic-diseases-ood\)](#)
- [OOD Divisions and Contact Information \(/about-fda/cder-offices-and-divisions/office-oncologic-diseases-ood-divisions-and-programs\)](#)
- [Information for Sponsors: Submissions to the Office of Oncologic Diseases \(OOD\) \(/about-fda/center-drug-evaluation-and-research-cder/information-sponsors-submissions-office-oncologic-diseases-ood\)](#)

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